

DOYLESTOWN, PA, USA

WMO: 725113

Lat: 40.330N

Lon: 75.123W

Elev: 394

StdP: 14.49

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 54786

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	9.5	13.6	-5.8	4.1	12.0	-1.1	5.2	16.5	23.1	28.4	20.5	26.9	7.6	320	0.429

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.8	91.1	73.9	88.7	72.8	86.1	71.4	76.7	86.1	75.3	83.8	74.1	81.9	7.2	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.9	128.1	80.1	72.8	123.4	78.8	71.9	119.7	78.0	40.3	86.2	38.9	84.1	37.8	81.7	86.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.8	16.5	14.0	4.5	95.5	4.9	3.5	1.0	97.9	-1.9	100.0	-4.7	101.9	-8.2	104.4
			3.0	79.4	4.8	2.2	-0.5	81.0	-3.3	82.3	-6.0	83.6	-9.5	85.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	53.4	30.3	32.7	40.6	52.1	61.6	70.3	75.2	73.5	67.0	55.4	44.8	35.8
	DBStd	17.53	10.29	9.32	9.36	8.82	7.64	6.23	4.84	5.11	6.52	8.14	8.35	8.84
	HDD50	2182	613	488	318	74	4	0	0	0	0	37	198	450
	HDD65	5198	1074	903	756	399	161	23	1	3	53	314	605	906
	CDD50	3423	3	5	26	138	364	608	783	729	510	205	43	9
	CDD65	963	0	0	0	12	56	181	318	266	113	17	0	0
	CDH74	8151	0	0	5	157	547	1564	2819	2130	824	104	1	0
	CDH80	2725	0	0	0	43	148	499	1077	709	232	17	0	0
Wind	WSAvg	5.3	6.5	6.6	6.8	6.2	4.8	4.5	4.0	3.7	3.9	4.8	5.6	6.1
Precipitation	PrecAvg	47.6	3.6	2.8	4.1	3.9	3.7	4.3	4.7	4.2	4.4	4.1	3.4	4.3
	PrecMax	65.8	6.6	6.3	7.8	10.2	6.8	9.6	10.2	13.0	10.9	11.0	5.7	10.2
	PrecMin	34.7	1.7	0.8	1.1	1.3	0.9	0.9	0.9	1.3	0.5	0.3	1.1	1.0
	PrecStd	7.6	1.4	1.3	1.7	1.9	1.5	2.2	1.9	2.5	2.8	2.7	1.3	1.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	62.7	65.6	72.8	85.2	88.5	92.0	95.5	93.5	90.4	81.8	70.8	64.4
		MCWB	56.7	56.7	58.6	64.2	70.5	74.2	75.6	74.6	72.8	68.7	60.9	58.9
	2%	DB	56.3	57.9	65.8	78.7	84.2	88.9	91.5	90.1	85.9	76.6	66.2	58.6
		MCWB	52.1	50.6	54.8	61.8	68.9	72.8	74.2	74.1	70.7	66.1	59.0	55.2
	5%	DB	49.6	51.7	60.9	72.8	80.6	85.6	89.0	86.8	82.2	72.6	62.6	54.1
		MCWB	44.4	44.4	50.7	58.8	66.4	70.9	73.1	72.6	68.6	64.0	55.3	50.1
	10%	DB	44.9	45.9	55.5	67.9	76.1	82.3	86.0	83.8	79.0	68.8	58.0	49.0
		MCWB	40.8	39.9	48.0	56.1	64.6	69.4	71.7	71.0	67.4	61.5	52.4	44.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	58.7	59.4	61.1	68.9	73.3	76.4	78.6	79.0	76.4	71.9	64.3	61.3
		MCDB	61.6	64.5	70.6	79.7	84.0	87.0	90.6	88.9	85.8	77.3	68.0	63.0
	2%	WB	52.4	52.1	57.0	64.7	71.0	74.8	76.9	76.7	74.1	69.1	60.8	55.8
		MCDB	55.1	57.8	62.9	74.3	80.7	84.4	86.9	85.0	80.2	74.6	64.7	57.8
	5%	WB	45.8	44.9	53.2	61.4	68.9	73.4	75.4	75.1	72.4	66.2	57.0	50.9
		MCDB	48.9	49.3	58.7	70.1	77.7	82.3	84.2	82.7	77.7	71.1	60.5	53.5
	10%	WB	41.2	40.6	48.5	58.0	66.7	71.7	74.1	73.6	70.4	62.9	53.7	45.4
		MCDB	44.8	45.4	55.5	66.2	73.9	79.8	82.2	80.2	75.8	66.9	56.9	47.8
Mean Daily Temperature Range	5% DB	MDBR	14.2	15.9	17.6	20.0	19.5	19.0	18.8	18.1	18.8	17.9	16.9	13.8
		MCDBR	20.9	23.2	24.6	26.6	24.4	22.4	21.9	20.9	22.6	21.9	21.3	18.9
		MCWBR	17.2	17.5	16.6	14.0	11.7	9.5	7.8	7.8	9.7	12.0	14.3	16.5
	5% WB	MCDBR	19.1	21.2	23.1	23.0	21.8	19.6	18.5	16.9	17.2	17.6	18.0	17.9
		MCWBR	17.3	18.0	17.9	14.1	11.5	9.8	7.8	7.5	9.3	11.7	14.8	17.7
Clear-Sky Solar Irradiance	taub		0.305	0.319	0.356	0.386	0.432	0.490	0.489	0.483	0.397	0.356	0.323	0.305
	taud		2.458	2.405	2.374	2.319	2.219	2.094	2.115	2.118	2.328	2.455	2.509	2.512
	Ebn at Noon		272	285	284	281	270	253	252	250	266	267	264	264
	Edh at Noon		25	30	35	39	44	50	49	48	36	29	24	22
All-Sky Solar Radiation	RadAvg		594	882	1164	1511	1680	1839	1885	1627	1354	955	674	509
	RadStd		52	96	92	141	195	154	92	105	158	101	61	53

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (69 neighbors)	+0.74	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+216	+122

Nomenclature: See separate page